

Supplementary results: non-reasoning baseline & prompt-level fixes

The Silent Tax — typo robustness of reasoning LLMs

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1 Reasoning vs. non-reasoning model

Answered-only accuracy of DeepSeek-R1-Distill-Qwen-7B (R1) vs. Qwen2.5-7B-Instruct (same family/size, no reasoning training), identical questions and prompts, 500 per config.

config	GSM8K		ARC-C		MATH-500	
	R1	Qwen	R1	Qwen	R1	Qwen
clean	92.4%	92.4%	83.1%	91.6%	95.2%	76.2%
typo25_real10	85.0%	87.8%	85.6%	92.2%	93.9%	75.5%
typo25_real40	87.0%	85.8%	85.0%	90.9%	93.4%	72.9%
typo25_real70	86.2%	85.4%	81.5%	90.3%	93.9%	73.7%
typo50_real10	79.6%	83.0%	81.2%	90.6%	92.8%	72.7%
typo50_real40	79.0%	79.6%	82.0%	88.8%	91.4%	72.3%
typo50_real70	72.8%	75.4%	82.3%	87.9%	91.4%	72.1%
typo75_real10	73.0%	73.6%	79.7%	87.4%	89.1%	68.0%
typo75_real40	66.8%	72.0%	76.4%	89.0%	88.1%	66.1%
typo75_real70	62.7%	65.6%	74.1%	83.9%	86.9%	69.3%

Table 1: Reasoning training does not confer typo robustness: on GSM8K both models start at 92.4% clean yet R1 degrades slightly more. The real-word effect appears in both models.

2 Prompt-level fixes (GSM8K, R1)

config	no-fix	rewrite	Δ	warn	Δ
clean	92.4%	91.4%	-1.0%	92.4%	+0.0%
typo25_real10	85.0%	83.6%	-1.4%	85.4%	+0.4%
typo25_real40	87.0%	82.4%	-4.6%	87.0%	+0.0%
typo25_real70	86.2%	80.0%	-6.2%	84.8%	-1.4%
typo50_real10	79.6%	76.2%	-3.4%	78.2%	-1.4%
typo50_real40	79.0%	71.6%	-7.4%	73.2%	-5.8%
typo50_real70	72.8%	68.0%	-4.8%	73.8%	+1.0%
typo75_real10	73.0%	65.8%	-7.2%	70.0%	-3.0%
typo75_real40	66.8%	62.6%	-4.2%	64.0%	-2.8%
typo75_real70	62.7%	56.2%	-6.5%	61.4%	-1.3%

Table 2: Neither fix recovers accuracy. *rewrite* (“first correct the typos, then solve”) hurts in every cell (mean -5.1 over typo configs): it shortens reasoning by $\sim 35\%$ and locks in early misreadings. *warn* (“the text may contain typos”) is null-to-slightly-negative (mean -1.6).